

### Definition

### Nilpotent

A linear operator  $T : V \rightarrow V$  is called **nilpotent** if  $T^k = T_0$  for some integer  $k \geq 1$ .

An  $n \times n$  matrix  $N$  is called **nilpotent** if  $N^k = 0_{n,n}$  for some  $k \geq 1$ .

### Example

The following matrices are nilpotent:

$$A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \quad B = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

### Proposition 1

Let  $N$  be a nilpotent linear operator on a finite-dimensional vector space  $V$ . Then the only eigenvalue of  $N$  is 0.

*Proof:* Since  $N$  is nilpotent, there is an integer  $k \geq 1$  such that  $N^k = T_0$ . Let  $\lambda$  be an eigenvalue of  $N$  with a corresponding eigenvector  $v$ .

Hence  $N(v) = \lambda v$ . By induction,  $N^k(v) = \lambda^k v$ . So  $\lambda^k v = \mathbf{0}$ , but since  $v$  is nonzero, it follows that  $\lambda = 0$ . ■

### Proposition 2

Let  $N$  be a linear operator on a vector space  $V$ . If for some integer  $m \geq 1$ ,

$$\ker(N^m) = \ker(N^{m+1}),$$

then

$$\ker(N^m) = \ker(N^{m+1}) = \ker(N^{m+2}) = \dots$$

*Proof:* For any integer  $k \geq 1$ , we know

$$\ker(N^k) \subseteq \ker(N^{k+1}) \subseteq \dots$$

Set  $m$  to the given integer such that  $\ker(N^m) = \ker(N^{m+1})$ .

Let  $v \in \ker(N^{m+2})$ , meaning  $N^{m+2}(v) = \mathbf{0}$ . Hence  $N^{m+1}(N(v)) = \mathbf{0}$ , so

$$N(v) \in \ker(N^{m+1}) = \ker(N^m).$$

So  $N^{m+1}(v) = N^m(N(v)) = \mathbf{0}$ , meaning  $v \in \ker(N^{m+1})$ .

Therefore,  $\ker(N^{m+2}) = \ker(N^{m+1})$ , and the rest follows by induction. ■

### Proposition 3

Let  $N$  be a linear operator on an  $n$ -dimensional vector space  $V$ . Then there is an integer  $m \leq n$  such that

$$\ker(N^m) = \ker(N^{m+1}) = \ker(N^{m+2}) = \dots$$

*Proof:* It is sufficient to show there is an integer  $m \geq 1$  such that

$$\ker(N^m) = \ker(N^{m+1}).$$

Then the result follows from Proposition 2.

To begin, we know

$$\ker(N) \subseteq \ker(N^2) \subseteq \dots$$

Hence,

$$\dim \ker(N) \leq \dim \ker(N^2) \leq \dots$$

However, since the null space of  $N^k$  for any  $k \geq 1$  is a subspace of  $V$ ,

$$\dim \ker(N^k) \leq n.$$

Thus, there must be a “stopping point” in this chain of inequalities. That is, for some integer  $m \geq 1$ ,

$$\dim \ker(N^m) = \dim \ker(N^{m+1}) = \dots$$

$$\text{where } \dim \ker(N^{m+k}) \leq n \text{ for any } k \geq 1.$$

Furthermore, this inequality chain cannot have more than  $n$  strictly increasing terms. Hence  $m \leq n$ . The result follows by Proposition 2. ■

### Proposition 4

Let  $N$  be a nilpotent operator on an  $n$ -dimensional vector space  $V$ . Then  $N^n = T_0$ .

*Proof:* By Proposition 3, there is an integer  $m \leq n$  such that

$$\ker(N^m) = \dots = \ker(N^n) = \dots$$

Since  $N$  is nilpotent, there is an integer  $k \geq 1$  such that  $N^k = T_0$ . So  $\ker(N^k) = V$ .

If  $k < m$ , then  $N^k = N^m = N^n = T_0$ , and we are done.

Otherwise, if  $k \geq m$ , then  $k = m + r$  for some  $r \geq 0$ , and

$$\ker(N^m) = \ker(N^{m+r}) = \ker(N^k) = V.$$

Hence  $N^m = N^n = T_0$ , as required. ■

### Corollary

Let  $A \in M_{n \times n}(F)$  such that  $A^k = 0_{n,n}$  for some  $k \geq 1$ . Then  $A^n = 0_{n,n}$ .

### Theorem

Let  $T$  be a linear operator on an  $n$ -dimensional vector space  $V$ . Then  $V$  can be written as

$$V = \ker(T^n) \oplus \text{range}(T^n).$$

*Proof:* By Proposition 2, we know  $\ker(T^n) = \ker(T^{n+1}) = \dots$ .

Let  $v \in \ker(T^n) \cap \text{range}(T^n)$ . Then  $T^n(v) = \mathbf{0}$  and  $v = T^n(u)$  for some  $u \in V$ . Hence

$$\mathbf{0} = T^n(v) = T^n(T^n(u)) = T^{2n}(u).$$

So  $u \in \ker(T^{2n}) = \ker(T^n)$ , meaning  $v = \mathbf{0}$ . Therefore,

$$\ker(T^n) \cap \text{range}(T^n) = \{\mathbf{0}\}.$$

Now, apply the Rank-Nullity Theorem to  $T^n$  to get

$$\dim V = \dim \ker(T^n) + \dim \text{range}(T^n).$$

The result follows. ■

### Definition

### Generalized Eigenvector

Let  $T$  be a linear operator on a finite-dimensional vector space  $V$ . A nonzero vector  $v \in V$  is called a **generalized eigenvector** of  $T$  corresponding to an eigenvalue  $\lambda$  if there is an integer  $k \geq 1$  such that

$$(T - \lambda I)^k(v) = \mathbf{0}.$$

Eigenvectors as we have discussed them up to this point are also generalized eigenvectors.

### Definition

### Generalized Eigenspace

Let  $T$  be a linear operator on a finite-dimensional vector space  $V$ . If  $\lambda$  is an eigenvalue of  $T$ , the **generalized eigenspace** corresponding  $\lambda$  is defined as the set of all generalized eigenvectors corresponding to  $\lambda$  plus the zero vector, i.e.

$$G_\lambda = \{v \neq \mathbf{0} \in V : (T - \lambda I)^k(v) = \mathbf{0} \text{ for some } k \geq 1\} \cup \{\mathbf{0}\}.$$